

General Rules of Probability

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Objectives

- Define independent events
- Determine whether two events are independent
- Apply the general addition rule
- Define conditional probability
- Compute conditional probabilities
- Apply the general multiplication rule

Probability Rules

Rule 1. The probability $P(A)$ of any event A satisfies $0 \leq P(A) \leq 1$.

Rule 2. If S is the sample space in a probability model, then $P(S) = 1$.

Rule 3. A and B are mutually exclusive if $A \cap B = \Phi$

Rule 4. If A and B are mutually exclusive, $P(A \cup B) = P(A) + P(B)$.

This is the **addition rule for mutually exclusive events**.

Rule 5. For any event A , $P(A \text{ does not occur } (A^c)) = 1 - P(A)$.

Examples

Q1: Which of the following
cant be a probability ?
0.5, 0.001, -0.1, .79 , 1.0009

Q2: IF $A = \{2, 3, 4, 5\}$
 $B = \{5, 6, 7\}$ and
 $C = \{9, 10\}$

Which of these are mutually
exclusive?

A and C , B and C

- Q3: If $P(A) = 0.3$ and $P(B) = 0.5$, A and B are ME. What is the probability for A or B to occur?

$$0.3 + 0.5 = 0.8$$

- If $P(A) = 0.3$ and $P(B) = 0.8$, are A and B mutually exclusive?
- No , because $0.3 + 0.8 = 1.1$ and it should be less than 1

Independent Events

If two events A and B do not influence each other, and if knowledge about one does not change the probability of the other, the events are said to be **independent** of each other.

Multiplication Rule for Independent Events

Two events A and B are **independent** if knowing that one occurs does not change the probability that the other occurs. If A and B are independent:

$$P(A \cap B) = P(A) \times P(B)$$

Example on Independence

- You flip a coin and get a heads *and* you flip a second coin and get a tails.

The two coins don't influence each other.

- The probability of raining today *and* the probability of a pop quiz;

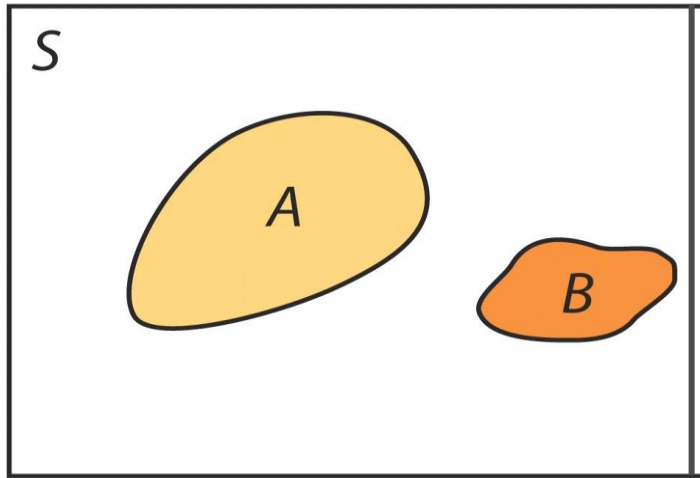
Quizzes happen, rain or shine.

Examples

If $P(A)=0.6$ and $P(B)=0.2$ A and B are independent find $P(A \cap B)$!
Then $P(A \cap B) = 0.2 \times 0.6 = 0.12$

If $P(A)=0.7$ and $P(B)=0.5$ and $P(A \cap B)=0.4$
Are A and B independent?
No because $P(A \cap B) \neq P(A) \times P(B)$

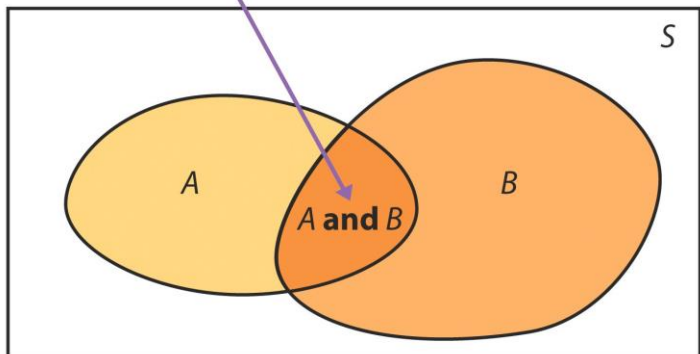
The General Addition Rule



We know if A and B are **mutually exclusive** events,

$$P(A \text{ or } B) = P(A) + P(B)$$

Outcomes here are double-counted by $P(A) + P(B)$.



Addition Rule for Any Two Events

For any two events A and B :

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Example

	Male	Female	total
Negative	10	12	22
Positive	13	5	18
Total	23	17	40

A = Male B = Negative

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A) = \frac{23}{40} , \quad P(B) = \frac{22}{40} \text{ and } P(A \cap B) = \frac{10}{40}$$

Then

$$P(A \cup B) = \frac{23}{40} + \frac{22}{40} - \frac{10}{40} = \frac{35}{40}$$

Conditional Probability

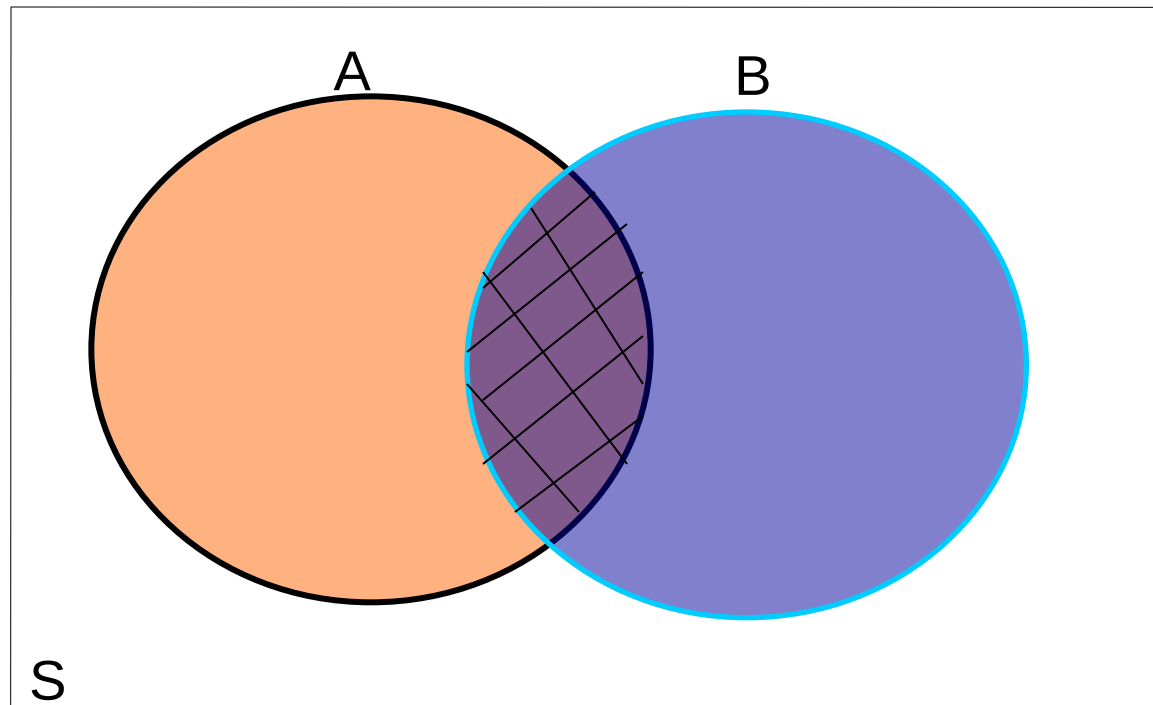
The probability that event A occurs if we know for certain that event B will occur is called **conditional probability**.

The conditional probability of A given B is denoted:

- **$P(A/B)$**
- **Read as the probability of A given B**

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



Example

	Male	Female	total
Negative	10	12	22
Positive	13	5	18
Total	23	17	40

A = Male B = Negative

$$P(A \cap B) = \frac{10}{40}$$

$$P(B) = \frac{22}{40}$$

Second method use the formula

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{10/40}{22/40} = \frac{10}{22}$$

Conditional Probability

Researchers asked people who exercise regularly whether they jog or walk. Fifty-eight percent of the respondents were male. Twenty percent of all respondents were males who said they jog. Find the probability that knowing he is a male, he is a jogger.

Relate: $P(\text{male}) = 58\%$

$P(\text{male and jogs}) = 20\%$

Define: Let $B = \text{male}$.

Let $A = \text{jogs}$.

Write: $P(A | B) = \frac{P(A \cap B)}{P(B)}$

$$\begin{aligned} &= \frac{0.2}{0.58} \\ &\approx 0.344 \end{aligned}$$

Example:

A number from the sample space $S = \{2, 3, 4, 5, 6, 7, 8, 9\}$ is randomly selected. Given the defined events A and B,

A: selected number is odd, and

B: selected number is a multiple of 3

find the following probabilities.

a) $P(B)$ b) $P(A \text{ and } B)$ c) $P(B/A)$

a) $B = \{3, 6, 9\}$ $P(B) = 3/8$

b) $P(A \text{ and } B) = P(\{3, 5, 7, 9\} \cap \{3, 6, 9\})$
 $= P(\{3, 9\}) = 2/8 = 1/4$

c) Probability of B given A has occurred:

$$P(B/A) = \frac{P(A \text{ and } B)}{P(A)} = \frac{1/4}{4/8} = 1/2$$

Example:

Given a family with two children, find the probability that both are boys, given that at least one is a boy.

$$S = \{gg, gb, bg, bb\}$$

$$A = \text{at least one boy} \quad A = \{gb, bg, bb\}$$

$$B = \text{both are boys} \quad B = \{bb\}$$

$$P(A \text{ and } B) = P(\{gb, bg, bb\} \cap \{bb\}) = P(\{bb\}) = 1/4$$

$$P(A) = P(\{gb, bg, bb\}) = 3/4$$

$$\frac{P(A \text{ and } B)}{P(A)} = \frac{1/4}{3/4} = 1/3$$

The General Multiplication Rule

The probability that events A and B both occur can be found using the **general multiplication rule**

$$P(A \text{ and } B) = P(A) \cdot P(B | A)$$

where $P(B | A)$ is the conditional probability that event B occurs given that event A has already occurred.

Note: Two events A and B that both have positive probability are **independent** if:

$$P(B|A) = P(B)$$

Multiplication Rule of Probability - Events are INDEPENDENT

Example:

A jar contains 4 red marbles, 3 blue marbles, and 2 yellow marbles. What is the probability that a red marble is selected and then a blue one with replacement?

$$\begin{aligned} P(\text{Red} \cap \text{Blue}) &= P(\text{Red}) \cdot P(\text{Blue}) \\ &= 4/9 \cdot 3/9 \\ &= 12/81 \\ &= 4/27 = 0.148 \end{aligned}$$

Multiplication Rule of Probability - Events are **DEPENDENT**

Example:

A jar contains 4 red marbles, 3 blue marbles, and 2 yellow marbles. What is the probability that a red marble is selected and then a blue one without replacement?

$$\begin{aligned} P(\text{Red} \cap \text{Blue}) &= P(\text{Red}) \cdot P(\text{Blue}/\text{Red}) \\ &= 4/9 \cdot 3/8 \\ &= 12/72 \\ &= 3/18 = 0.166667 \end{aligned}$$

Example

- Let $P(A)=.3$, $P(B)=.6$ $P(A/B)=.333$

FIND $P(A \cup B)$

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

We need to find

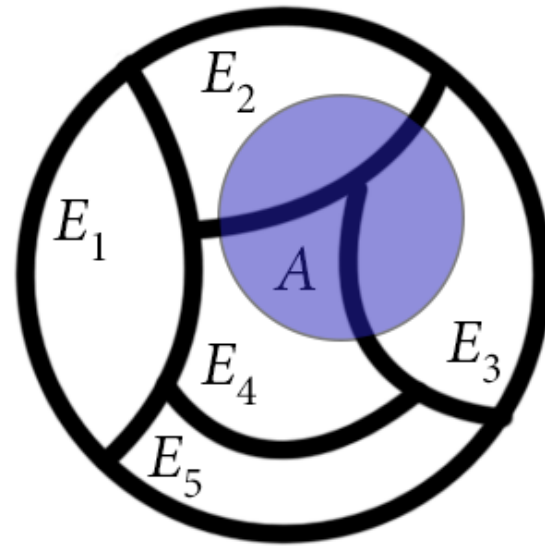
- $P(A \cap B) = P(A/B)P(B) = .6 \times .333 = .2$

Then

- $P(A \cup B) = .3 + .6 - .2 = .7$

Total Probability

- Total probability for event



$$P(A) = P(A|E_1)P(E_1) + P(A|E_2)P(E_2) + \cdots + P(A|E_k)P(E_k)$$

- Where $E_1, E_2, \dots, E_k$ are k mutually exclusive and collectively exhaustive events

Example

- $P(\text{long hair/boy})=.02$
- $P(\text{long hair/girl})=.76$
- $P(\text{boy})=.5$
- Find the probability for having long hair

$$P(\text{long hair})= P(\text{long hair/boy})P(\text{boy}) + P(\text{long hair/girl})P(\text{girl})$$

$$=.02 \times .5 + .76 \times .5 = .39$$